

ElectroWire



LIGHTNINGWIRE™ TRANSSEPTAL PUNCTURE SYSTEM INSTRUCTIONS FOR USE

Failure to carefully read, understand, and follow all instructions (including all contraindications, warnings, and precautions) may result in injury or death to the patient and/or healthcare providers.

Caution: Federal (U.S.A.) law restricts this device to sale by or on the order of a physician.



I. Device Description

The LightningWire™ Transseptal Puncture System (TPS) is a needle-free, powered, wire-based TPS that is used to create an inter-atrial septal defect, enabling transseptal access to the left atrium. It consists of a LightningWire with proprietary hydrophobic coating (**Figure 1**) and an Activation Cable (**Figure 2**). It is a monopolar transcatheter electrosurgical accessory, wherein the tip of the LightningWire is the active electrode. The LightningWire is designed to be inserted through a commercially available transseptal introducer set and features visible markers (**Figure 1**) along its length to assist with aligning the wire tip in a compatible transseptal sheath and/or dilator assembly. The LightningWire is provided with a length of 180 cm or 260 cm (**Table 1**). The LightningWire is reversibly connected to a commercially available electrosurgical generator (ESG) by the Activation Cable, which includes an Activation Button that controls the flow of radiofrequency (RF) energy. The length of the Activation Cable is 3 meters. The LightningWire TPS is provided sterile and is for single use.

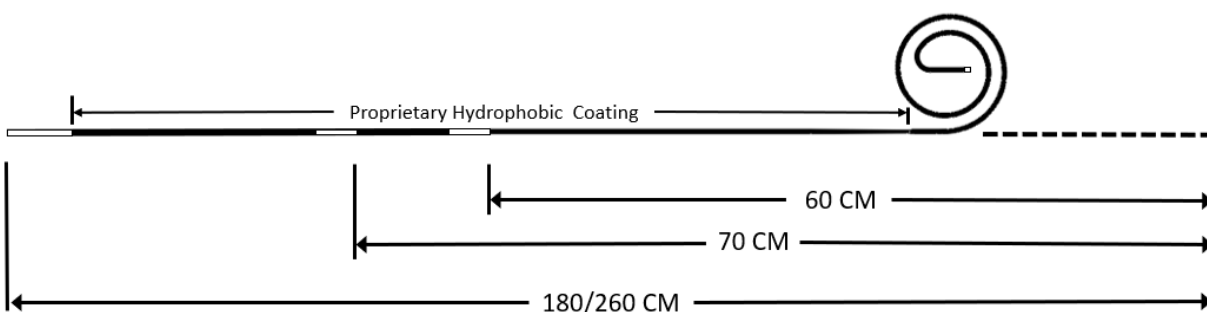


Figure 1: LightningWire

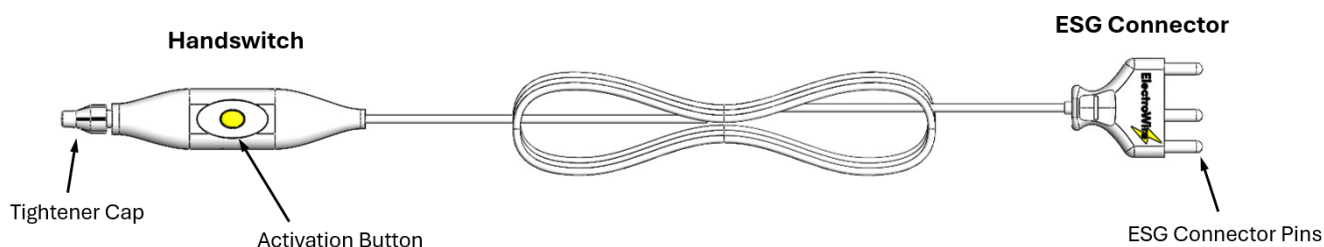


Figure 2: LightningWire Activation Cable

Table 1. Available TPS Models

Model No.	Length	Support Level
01-180	180 cm	Extra Support (ES)
01-260	260 cm	Extra Support (ES)



II. Indications for Use

The LightningWire TPS is used to create an atrial septal defect in the heart, whereby various cardiovascular catheters are introduced.

III. Contraindications

The LightningWire TPS is not recommended for use with:

- Any conditions that do not require the creation of an atrial septal defect
- Known or suspected atrial thrombus
- Known or suspected atrial myxoma
- Cerebrovascular accident (CVA) within one week
- Any indication of increased risk of life-threatening bleeding with anticoagulation
- Any inability to tolerate anticoagulation for any reason
- History of inter-atrial septal patch repair
- History of device closure of atrial septal defect or patent foramen ovale

IV. Warnings

- Only physicians with appropriate training, a thorough understanding of fluoroscopically and echocardiographically guided transseptal puncture, and a thorough understanding of electrosurgery should use this device.
- The LightningWire TPS is supplied STERILE using an ethylene oxide process. Do not use if the package appears to have been opened or is damaged.
- If fluoroscopy is used, patients and healthcare providers may be exposed to ionizing radiation during inter-atrial septal puncture procedures. This exposure can result in acute radiation injury as well as increased risk for stochastic and deterministic effects. Fluoroscopy use should be managed according to the ALARA (“as low as reasonably achievable”) principle.
- RF energy should be applied only under the immediate control of a physician who may be expected to interrupt RF energy application at the first sign of unexpected results.
- The patient’s heart rhythm and hemodynamic parameters should be monitored during the procedure.
- The LightningWire TPS should not be altered in any way. Doing so may make its use unsafe.
- The LightningWire TPS is intended for single patient use only. Do not attempt re-sterilization or reuse. Reuse can cause patient injury and/or the communication of infectious disease(s) from one patient to another. Reuse may result in patient complications.
- The LightningWire TPS should be used only with ESGs having a maximum open circuit peak-to-peak voltage (V_{p-p}) output of 3000 volts or less in the “pure cut” (continuous sinusoidal) mode. The LightningWire TPS is a Type CF device and rated for 623 volts maximum.
- The LightningWire TPS should be used only with ESGs having a monopolar accessory port configured for use with industry-standard electrosurgical pencils.
- The LightningWire TPS should be used only with defibrillation-proof ESGs.



- The LightningWire must be used with the Activation Cable provided. Attempts to use it with other connector cables can result in injury to the patient and/or operator.
- The LightningWire must be used with commercially available 0.032" - 0.035" compatible transseptal introducer sets, which should be used according to the manufacturer's instructions. Use of incompatible accessory devices may damage the LightningWire and/or its coating, may damage the accessory devices, and may cause injury to the patient and/or operator.
- The distal tip and distal curve of the LightningWire are fragile. Be careful not to damage the distal tip or the distal curve while handling the LightningWire. If the distal tip or the distal curve is damaged at any time during use, discard the LightningWire immediately. Do not attempt to straighten the distal tip if bent. Do not attempt to straighten or reshape the preformed pigtail shape. Damage to the LightningWire can lead to injury to the patient and/or operator.
- Failure to abide by the warnings in this labeling might result in damage to the LightningWire hydrophobic coating, which may necessitate intervention or result in serious adverse events.
- The safety and effectiveness of the LightningWire TPS have not been established or is unknown in vascular regions other than those specifically indicated.
- The LightningWire TPS is not intended for use in children or in pregnant or nursing women.
- Do not attempt to insert or retract the LightningWire through a metal cannula or a percutaneous needle, which may damage the device and its coating and lead to injury to the patient and/or operator.
- Do not attempt to snare the LightningWire, which may damage the device and/or its coating, and lead to injury to the patient and/or operator.
- Do not press the Activation Button when the tip of the LightningWire is proximal to the tip of the transseptal introducer set's dilator.
- Do not press the Activation Button when the tip of the LightningWire is more than 5 mm beyond the tip of the transseptal introducer set's dilator.
- Do not use "cut" or "coag" power settings other than those described in the Directions for Use in **Section X** below.

V. Precautions

- Do not attempt to use the LightningWire TPS before thoroughly reading the Instructions for Use.
- Following the instructions in this document is required in order to mitigate the risks and potential complications associated with transseptal access procedures, including, but not limited to, embolism and perforation.
- RF puncture procedures should be performed only by physicians thoroughly trained in the techniques of RF-powered puncture in a fully equipped catheterization laboratory.
- The sterile packaging should be visually inspected prior to use to detect any compromise. Ensure that the packaging has not been damaged. Do not use the device if the packaging has been compromised.
- Visually inspect the LightningWire TPS prior to use. Do not use any damaged or defective devices.
- Do not use the LightningWire TPS after the use-by date indicated on the label.
- The LightningWire TPS is intended for use only with compatible transseptal introducer sets and ESGs as described in **Section VII**.

- Read, understand, and follow the manufacturer's Instructions for Use for the transseptal introducer set that will be used with the LightningWire TPS.
- Read, understand, and follow the manufacturer's Instructions for Use for the ESG that will be used with the LightningWire TPS.
- Read, understand, and follow the manufacturer's Instructions for Use for the patient return electrode that will be used with the LightningWire TPS.
- Always use patient return electrodes that meet or exceed IEC 60601-2-2 requirements and are compatible with the ESG that will be used with the LightningWire TPS.
- In order to prevent the risk of ignition, ensure that flammable materials are not present in the room during RF energy application.
- Do not attempt to insert and use the proximal end of the LightningWire as the active tip.
- Do not excessively bend the LightningWire. Excessive bending or kinking of the LightningWire may damage essential components, including the coating, and may cause patient injury. Care must be taken when handling the LightningWire.
- Careful manipulation of the LightningWire must be performed to avoid vessel trauma. If resistance is encountered, DO NOT use excessive force to advance or withdraw the LightningWire or ancillary sheath and/or dilator assemblies. Excessive force may lead to bending, or kinking of the LightningWire, limiting advancement and retraction of ancillary devices and potentially leading to patient complications.
- LightningWire and ancillary device advancement should be done under imaging guidance. The visible markers on the wire body are only an approximate guide for use in positioning the LightningWire's tip near the distal end of the dilator.
- Care should be taken to avoid interaction between the LightningWire and intra-cardiac devices, such as pacemaker or defibrillator leads. Such interaction may result in lead dislodgement and related complications.
- Do not attempt to deliver RF energy until the active tip of the LightningWire is confirmed to be in good contact with the target tissue on the inter-atrial septum.
- Avoid RF energy delivery to the LightningWire with incompatible devices, which may lead to thermal injury, ineffective puncture, or failure to puncture.
- Do not exceed five (5) RF energy applications to a single LightningWire.
- Never disconnect the Activation Cable from the ESG while RF energy is being delivered.
- Never disconnect the Activation Cable from the ESG by pulling on the cable. Failure to disconnect the cable properly may result in damage to the ESG and/or cable.
- The ESG is capable of delivering significant electrical power. Patient or operator injury can result from improper handling of the LightningWire and/or patient return electrode, particularly when operating the device.
- Placement of the patient return electrode on the thigh could be associated with higher impedance.
- During RF energy application, the patient should not be allowed to come in contact with grounded metal surfaces.
- Apparent low power output or failure of the equipment to function properly at normal settings may indicate faulty application of the patient return electrode, failure of an electrical lead, or poor tissue contact at the active tip. Check for obvious equipment defects or misapplication.



Attempt to better position the distal tip of the LightningWire against the inter-atrial septum. Only increase the power setting if low power output persists despite other corrective measures.

- Electroanatomical mapping (EAM) guidance with the LightningWire TPS has not been studied.
- Prior to using the LightningWire TPS, physicians should assess and communicate its patient-specific risks and benefits.
- For inter-atrial septal puncture with the LightningWire TPS, certain patient conditions require special consideration. Such conditions include, but aren't limited to, an enlarged aortic root, a small left atrium, a dilated right atrium, fibrotic or calcified inter-atrial septal tissues, the presence of atrial septal defect or patent foramen ovale closure devices, the presence of intra-cardiac devices such as pacemaker/defibrillator leads, distortion of normal thoracic anatomy, and congenital malformations of the heart.

VI. Potential Complications

Adverse events that may occur while creating an atrial septal defect include:

Allergic reaction	Sepsis/Infection	Thromboembolic events
Vessel perforation	Atrial Flutter or Fibrillation	Myocardial Infarction
Vessel spasm	Ventricular arrhythmias	Stroke
Hemorrhage	Vascular thrombosis	Perforation of the myocardium
Hematoma	Arteriovenous fistula	Ventricular Tachycardia
Pain and/or Tenderness	Vascular Trauma	Pericardial effusion
Tachycardia	Foreign body/wire fracture	Pericardial tamponade
Wire entrapment/entanglement	Thermal injury	Additional Surgical Procedure

Consult the respective manufacturer's labeling for potential complications associated with the use of ancillary devices that may be used in conjunction with the LightningWire TPS.

VII. Ancillary Devices Required But Not Provided

Transseptal puncture procedures should be performed in a specialized clinical setting equipped with appropriate imaging, physiologic monitoring, and emergency equipment. The following ancillary devices are required and are commercially available from other manufacturers:

- Conventional percutaneous needle for initiating vascular access
- 0.032" - 0.035" compatible transseptal introducer set (sheath and dilator), such as the Agilis NxT and Fast-Cath SL1 that were subject to compatibility testing with the LightningWire TPS.
- ESG with maximum open circuit peak-to-peak voltage (V_{p-p}) output of no more than 3000 volts in the "pure cut" (continuous sinusoidal) mode and having a monopolar accessory port configured for use with industry-standard electrosurgical pencils. The LightningWire TPS is a Type CF device rated for 623 volts maximum and should be used only with defibrillation-proof ESGs. The Megadyne Megapower 1000 was subject to compatibility testing with the LightningWire TPS.
- Patient return electrode meeting or exceeding IEC 60601-2-2 requirements



VIII. Inspection Prior to Use

Prior to use, the LightningWire TPS should be carefully examined for damage or defects, as should all equipment used in the procedure. Do not use defective devices or equipment.

The LightningWire TPS is provided in a cardboard shelf box, which contains a thermoformed tray within a Tyvek™ pouch. The pouch should be examined for obvious damage. Do not use any product with damaged packaging.

The LightningWire TPS is labeled on both the cardboard shelf box and the Tyvek™ pouch. The contents of the package are clearly identified on both labels, along with other information concerning the contents of the package. Careful review of the label is required to ensure the proper device has been selected for the procedure and that the product is not expired. Do not use expired product.

IX. Storage and Handling

The LightningWire TPS should be stored in a cool, dry place. Keep out of sunlight.

The used product should be disposed of in accordance with the facility's standard for solid and sharps biohazard waste procedures.

X. Directions for Use

- All instructions for equipment required should be carefully read, understood, and followed. Failure to do so may result in complications.
- The LightningWire TPS is supplied sterile. Use aseptic technique when opening the package and handling the devices in the sterile field.
- Place the commercially available patient return electrode on the patient, and ensure it is connected to a compatible ESG according to the manufacturer's instructions.
- Connect the Activation Cable's ESG connector to the compatible ESG's monopolar accessory port by aligning the ESG connector pins with the holes in the monopolar accessory port and pushing until the ESG connector fits firmly into the socket. Do not use excessive force, as doing so may damage the ESG connector (including its pins), the ESG connector cable, and/or the ESG.
- Adjust the power settings of the compatible ESG to provide 15 Watts of "cut" and 0 Watts (or minimum setting allowed) of "coag", which will yield a "pure cut" (continuous sinusoidal) RF waveform.
- The LightningWire is provided in a dispenser coil with a tip straightener. Gently pull the LightningWire and tip straightener out of the dispenser coil together. The LightningWire is ready for use.
- Prepare the commercially available transseptal introducer set according to the manufacturer's instructions. Flush the wire lumen of the dilator with sterile fluid.
- Perform a standard femoral vein puncture using an access needle (not supplied)
- Insert a transseptal introducer set through the access site, advancing over a guidewire into the superior vena cava (SVC) using imaging guidance.
- Remove the guidewire and exchange for the LightningWire. Use the provided tip straightener to insert the LightningWire into the transseptal introducer set's dilator.
- Using imaging guidance, position the tip of the LightningWire at the tip of the transseptal introducer set's dilator. Visible markers on the LightningWire can be used to facilitate positioning of the wire tip near the distal end of the dilator.

- Carefully insert the proximal end of the LightningWire into the distal tip of the Activation Cable's Handswitch. Stop when resistance is met, and tighten the tightener cap to secure the proximal end of the LightningWire within the Handswitch. Gently tug the LightningWire to confirm a secure connection.
- Using standard transseptal puncture technique and appropriate imaging guidance, withdraw the transseptal assembly (introducer set and LightningWire) into the right atrium, and place the tip of the transseptal introducer set's dilator against the fossa ovalis (FO). Appropriate imaging guidance may include fluoroscopy and/or echocardiography.
- The transseptal introducer set's dilator should be visualized tenting the septum at the FO. Using imaging guidance, visualize the exact location of tenting of the FO to ensure that tenting is localized to the intended inter-atrial septal puncture site. According to individual patient anatomy, adjustment of the introducer set's curvature and/or the use of a different introducer set may be needed to achieve appropriate tenting. Follow the introducer set manufacturer's Instructions for Use.
- Using imaging guidance and tactile feedback, advance the LightningWire so that the distal tip is engaging the septum at the FO but not extending more than 5 mm beyond the tip of the transseptal introducer set's dilator.
- Once appropriate positioning has been achieved, RF energy may be delivered by pressing the Activation Button. Watch for the tip of the LightningWire to cross the inter-atrial septum and/or for microbubbles to appear in the left atrium, and release the activation button to stop RF energy delivery as soon as either is seen. The required duration of RF energy delivery may be less than one second.
- With adequate tenting of the FO, the LightningWire may cross passively (without any advancement of the guidewire) as soon as RF energy is delivered. During application of RF energy, carefully advance the LightningWire only as far as needed to cross the inter-atrial septum. The LightningWire should not be advanced more than 5 mm beyond the tip of the transseptal introducer set's dilator with the Activation Button depressed.
- If the LightningWire fails to cross the inter-atrial septum after a maximum of 3 seconds of RF energy delivery, release the activation button to stop RF energy delivery. The LightningWire should be pulled back into the introducer set's dilator. A new attempt at transseptal puncture may be made. Increasing power in increments of 5W (up to 25W) may be helpful.
- If the inter-atrial septum has not been crossed successfully after five attempts, the LightningWire should be discarded. An alternative means of inter-atrial septal puncture may be considered.
- Once the inter-atrial septal puncture is successfully completed, the LightningWire should be mechanically advanced without any further RF energy application. LightningWire positioning in the left atrium is sufficient when the full pigtail shape of the LightningWire has crossed the inter-atrial septum and is observed in the left atrium using fluoroscopy and/or echocardiography.
- If desired, advance the transseptal introducer set over the LightningWire into the left atrium. If desired, the LightningWire and transseptal introducer set's dilator may be removed together, leaving the transseptal introducer sheath in the left atrium.
- To remove the Activation Cable from the LightningWire, loosen the tightener cap to release the LightningWire. Gently remove the Activation Cable from the proximal end of the LightningWire. The Activation Cable may be set aside.
- If desired, remove the transseptal introducer set over the LightningWire. Take care to maintain the distal tip position of the LightningWire in the left atrium. The LightningWire may be used to deliver diagnostic and/or therapeutic devices to the left atrium.



- When use is complete, withdraw the LightningWire slowly. If any resistance is felt, imaging guidance should be used to correct any impediment to withdrawal.

Table 2: ESG Settings and RF Energy Application Parameters for the LightningWire TPS

Cut Power	15-25 W
Coag Power	0 W (or minimum setting allowed)
RF Energy Application Time (per application)	Less than 1 s Up to maximum of 3 s
Maximum Number of RF Energy Applications	5

XI. Troubleshooting

Problem	Comments	Troubleshooting
LightningWire's proximal end cannot be inserted into Handswitch.	The tightener cap, when too tight, may prevent insertion of the LightningWire into the Handswitch.	Check that the tightener cap is not too tight.
ESG connector does not fit into ESG.	The ESG connector will fit only into the monopolar accessory port of compatible ESGs.	Check that the ESG is compatible, that the monopolar accessory port has been correctly identified, and that the ESG connector pins are properly aligned.
ESG displays an error message.	Compatible ESGs may display error messages for various reasons.	Consult the ESG manufacturer's Instructions for Use.
RF energy is not being delivered.	Secure connection of the LightningWire, Activation Cable, and ESG is needed for RF energy delivery.	Check that the ESG connector is securely inserted at the ESG's monopolar accessory port, and check that the LightningWire is fully inserted and secured within the Handswitch.
LightningWire breaks or kinks.	Breaks and/or kinks are a potential cause of patient and/or operator injury.	Discard and replace the LightningWire.

XII. Customer Service and Product Return Information

If you have any problems with or questions about the LightningWire TPS, contact LightningWire technical support personnel.

1. In order to return any product, you must have a return authorization number before shipping the product back to LightningWire.
2. ElectroWire Medical will not accept any used devices without proper decontamination being performed under the guidelines of the hospital where the device was used. Ensure that any

product being returned to ElectroWire Medical has been cleaned and decontaminated in compliance with any local or organization guidelines.

3. Contact Information:

ElectroWire Medical
 143 Industrial Dr.
 Lexington, SC 29072 USA
 (800) 654-8238
info@electrowiremedical.com
www.electrowiremedical.com

XIII. Labeling and Symbols

Symbol	Symbol Title	Explanatory Text	ISO/IEC Reference	Reference Clause
	Manufacturer	Indicates the medical device manufacturer.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.1.1
	Date of Manufacture	Indicates the date when the medical device was manufactured	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.1.3
	Do not re-sterilize	Indicates a medical device that is not to be re-sterilized.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.2.6
	Do not reuse	Indicates a medical device that is intended for one use or for use on a single patient during a single procedure.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.4.2
	Catalog or Model number	Indicates the manufacturer's catalog number so that the medical device can be identified.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.1.6
	Batch code	Indicates the manufacturer's batch code so that the batch or lot can be identified.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.1.5

Symbol	Symbol Title	Explanatory Text	ISO/IEC Reference	Reference Clause
	Prescription only	Caution: Federal (USA) law restricts this device to sale by, or on the order of, a physician.	21 CFR 801.109	N/A
	Do not use if package is damaged	Indicates that the medical device must not be used if the package holding the device is damaged.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.2.8
	Sterilized using ethylene oxide	Indicates a medical device that has been sterilized using ethylene oxide.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.2.3
	Consult instructions for use	Indicates the need for the user to consult the instructions for use.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.4.3
	Use by date	Indicates the date after which the medical device is not to be used. Date format is YYYY-MM-DD	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.1.4
	Keep dry	Indicates a medical device that needs to be protected from moisture.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.3.4
	Keep away from sunlight	Indicates a medical device that needs to be protected from sunlight.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.3.2
	Temperature limit	Indicates the temperature limits to which the medical device can be safely exposed.	ISO & ANSI/AAMI 15223.1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.3.7
	Contains no natural rubber latex	Indicates the absence of dry natural rubber or natural rubber latex	ISO & ANSI/AAMI 15223.1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.4.5

Symbol	Symbol Title	Explanatory Text	ISO/IEC Reference	Reference Clause
	Non-pyrogenic	Indicates a medical device that is non-pyrogenic.	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.6.3
	Unique device identifier	Indicates a carrier that contains unique device identifier information	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.7.10
	Medical device	Medical device	ISO & ANSI/AAMI 15223-1 Medical Devices – Symbols to be used with medical device labels- General requirements.	5.7.7
	MR Unsafe	Known to pose hazards in all MR environments	ASTM F2503-13 Standard Practice for Marking Medical Devices and Other Items for Safety in the Magnetic Resonance Environment.	7.3.3
	Packaging unit	Indicates the number of pieces in the package	ISO 7000 — Graphical symbols for use on equipment.	2794